

Contents

Functions and Continuity	2
Monday, April 13 - Introduction to Differentiability	2
Wednesday, April 15 - Non-Differentiable Functions	6
Monday, April 20 - The Intermediate Value Prop. and Darboux's Theorem	10
Wednesday, April 22 - Mean Value Theorem	13
Friday, April 24 - L'Hospital's Rule	17

Understanding Analysis Sections Covered: Section 5.2

Definitions and Theorems to include in Analysis Journal:

- Definition 5.2.1 (differentiability)
- Theorem 5.2.3 (differentiable $\Rightarrow$ continuous)
- Theorem 5.2.4 (algebraic differentiability thm)
- Theorem 5.2.5 (chain rule)

Differentiability: Intuition

Differentiability

Let $g : A \rightarrow \mathbb{R}$ be defined on an interval A . Given $c \in A$, the _____ of g at c is defined by

$$g'(c) =$$

provided this limit exists. In this case, we say g is _____ at c . If $g'(c)$ exists for all points $c \in A$, then we say g is _____.

1. *Discuss:*

(a) Have you seen other definitions of the derivative?

(b) Why does A need to be an interval?

(c) How does the definition of differentiability compare to the definition of continuity?

Theorem 5.2.3 (Differentiability $\Rightarrow$ Continuity)

If $g : A \rightarrow \mathbb{R}$ is differentiable at a point $c \in A$, then g is continuous at c as well.

Scratch Work:

Proof.

□

2. Prove that $\frac{d}{dx}x^n = nx^{n-1}$ for any positive integer n .

Proof.

□

Theorem 5.2.5 (Algebraic Differentiability Theorem)

Let f and g be functions defined on an interval A , and assume both are differentiable at some point $c \in A$. Then,

(i) $(f + g)'(c) =$

(ii) $(kf)'(c) =$

(iii) $(fg)'(c) =$

(iv) $(f/g)'(c) =$

3. Prove part (iii) of the Algebraic Differentiability Theorem (product rule).

Proof.

□

Theorem 5.2.5 (Chain Rule)

Let $f : A \rightarrow \mathbb{R}$ and $g : B \rightarrow \mathbb{R}$ satisfy _____ so that $g \circ f$ is defined on A . If f is differentiable at $c \in A$ and if g is differentiable at _____, then $g \circ f$ is differentiable at c and

$$(g \circ f)'(c) =$$

“Dirty” Proof:

After Class:

Pick & Prove Problems:

★★ Exercise 5.2.1

Supply proofs of parts (i) and (ii) of the Algebraic Differentiability Theorem (sum rule and constant multiple rule).

★★ Exercise 5.2.3

- (a) Use the definition of the derivative to find the derivative of $h(x) = 1/x$.
- (b) Combine the result in part (a) with the Chain Rule and Product Rule to furnish a proof of the Quotient Rule.
- (c) Supply a direct proof of the Quotient Rule by algebraically manipulating the difference quotient for f/g in a style similar to how we proved the Product Rule.

★★ Exercise 5.2.5

Let $f_a(x) = \begin{cases} x^a & \text{if } x > 0 \\ 0 & \text{if } x \leq 0. \end{cases}$

- (a) For which values of $a \in \mathbb{R}$ is f continuous at zero? Use any definition/theorems you wish.
- (b) For which values of $a \in \mathbb{R}$ is f differentiable at zero? Use the definition of differentiability.
- (c) For which values of $a \in \mathbb{R}$ is f twice-differentiable (derivative is differentiable) at zero? You may use the fact that $\frac{d}{dx}x^a = ax^{a-1}$ even though we haven't proven it for all $a \in \mathbb{R}$.

★★ Exercises 5.2.6

Let g be defined on an interval A , and let $c \in A$.

- (a) Explain why $g'(c)$ could have been given by

$$g'(c) = \lim_{h \rightarrow 0} \frac{g(c+h) - g(c)}{h}.$$

- (b) Assume A is open. If g is differentiable at $c \in A$, show

$$g'(c) = \lim_{h \rightarrow 0} \frac{g(c+h) - g(c-h)}{2h}.$$

Why does A have to be an open interval in this case?

Understanding Analysis Sections Covered: 5.1, 5.2

Definitions and Theorems to include in Analysis Journal: None

Before Class:

Readings: Alcock, Sections 8.1-8.3, 8.5

Analysis Journal Prompts:

- How is differentiability different than the idea of differentiation?
- What are the different ways a function could be non-differentiable?
- Consider the modified topologist's sine function $f : \mathbb{R} \rightarrow \mathbb{R}$ given by

$$f = \begin{cases} x \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}.$$

Is this function continuous at 0? Is this function differentiable at 0?

For each example, discuss whether it is differentiable at 0. If it is not, describe what “kind” of non-differentiability we have. Also, discuss what happens to $f'(x)$ as $x \rightarrow 0$.

Example 1: Topologist's Sine Function: $f : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ given by $f(x) = \begin{cases} \sin(\frac{1}{x}) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$

Example 2: $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = \begin{cases} x \sin(\frac{1}{x}) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$

Example 3: $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = \begin{cases} x^2 \sin(\frac{1}{x}) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$

Observation: *Derivatives* are not necessarily continuous.

Example 4: $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = \begin{cases} x^{1.5} \sin(\frac{1}{x}) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$

After Class:

Pick & Prove Problems:

★★ Supplemental Exercise

Consider functions of the form

$$g_a(x) = \begin{cases} x^a \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}.$$

- (a) Prove that for all $a > 0 \in \mathbb{R}$, $g_a(x)$ is continuous everywhere in $\mathbb{R}$.
- (b) Prove that for all $a > 1 \in \mathbb{R}$, $g_a(x)$ is differentiable everywhere in $\mathbb{R}$.
- (c) Prove that for all $a > 3 \in \mathbb{R}$, $g_a(x)$ is twice-differentiable everywhere in $\mathbb{R}$.

★ Exercise 5.2.7 (Modified)

For the following problem, you may use the results from the problem above without proof. Provide short justifications for your answers but not a full proof.

Consider functions of the form

$$g_a(x) = \begin{cases} x^a \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}.$$

Find a range of real-number values for a such that...

- (a) ... g_a is differentiable on $\mathbb{R}$ but g'_a is unbounded on $[0, 1]$.
- (b) ... g_a is differentiable on $\mathbb{R}$ and g'_a is continuous but not differentiable at 0.
- (c) ... g_a is differentiable on $\mathbb{R}$ and g'_a is differentiable on $\mathbb{R}$, but g''_a is not continuous at 0.

★★★ Exercise 5.2.10 (Slightly Modified)

A familiar mantra from calculus is that a differentiable function is increasing if its derivative is positive, but this statement requires some sharpening in order to be completely accurate.

- (a) Show that the function

$$g(x) = \begin{cases} x/2 + x^2 \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

is differentiable on $\mathbb{R}$ and satisfies $g'(0) > 0$.

Now prove that g is not increasing over any open interval containing 0, that is, show for any $\varepsilon > 0$, there exist $x, y \in V_\varepsilon(0)$ such that $x < y$ but $f(x) \geq f(y)$.

- (b) Why doesn't this work for the similar function below?

$$g(x) = \begin{cases} x + x^2 \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Understanding Analysis Sections Covered: 5.2

Definitions and Theorems to include in Analysis Journal:

- Theorem 5.2.6 (Interior Extremum Theorem)
- Theorem 5.2.7 (Darboux's Theorem)

Recall: Intermediate Value Theorem

Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. If L is a real number satisfying $f(a) < L < f(b)$ or $f(a) > L > f(b)$, then...

Recall: Intermediate Value Property

A function f has the **intermediate value property** on a closed interval $[a, b]$ if for all $x < y$ in $[a, b]$ and all L between $f(x)$ and $f(y)$, then...

Recall: If a function has the intermediate value property on a closed interval, is it continuous on that interval? Not necessarily.

Example:

From last time: Derivatives are not necessarily continuous.

Example:

Darboux's Theorem

So, what kinds of discontinuities can a derivative have?

Theorem 5.2.6 (Interior Extremum Theorem)

Let f be differentiable on an open interval (a, b) . If f attains a maximum or a minimum value at some point $c \in (a, b)$, then $f'(c) = \underline{\hspace{2cm}}$.

Proof.

□

Theorem 5.2.7 (Darboux's Theorem)

If f is differentiable on an interval $[a, b]$ and if α is strictly between $f'(a)$ and $f'(b)$, then...

Proof.

□

After Class:

Pick & Prove Problems:

★ Exercise 5.2.9

Decide whether each statement is true or false. Justify your answer for those that are true and find a counterexample for those that are false.

- (a) If f' exists on an interval and is not constant, then f' must take on some irrational values.
- (b) If f' exists on an open interval and there is some point c where $f'(c) > 0$, then there exists a δ -neighborhood around c in which $f'(x) > 0$ for all $x \in V_\delta(c)$.
- (c) If f is differentiable on an interval containing 0 and $\lim_{x \rightarrow 0} f'(x) = L$, then it must be that $L = f'(0)$.

★★ Supplemental Exercise

Exercise 5.2.9(b) is false. Make it a true statement by inserting a three-word phrase, and prove the resulting statement.

★★ Exercise 5.2.11a (Modified)

Assume that g is differentiable on $[a, b]$ and satisfies $g'(a) < 0$. Show that there exists a point $x \in (a, b)$ where $g(a) > g(x)$, i.e. $g(a)$ cannot be a minimum value on $[a, b]$.

Hint: Follow the style of the proof of the Interior Extremum Theorem. (Note that I did it a little differently in class than it's written in your book but the ideas are the same.)

Understanding Analysis Sections Covered: 5.3

Definitions and Theorems to include in Analysis Journal:

- Theorem 5.3.1 (Rolle's Theorem)
- Theorem 5.3.2 (Mean Value Theorem)
- Theorem 5.3.3 (zero derivative implies constant function)
- Theorem 5.3.4 (functions with same derivative differ only by a constant)
- Theorem 5.3.5 (Generalized MVT)

Before Class:

Readings: None.

Analysis Journal Prompts:

- Fill in the Mean Value Theorem below from your textbook.
The Mean Value Theorem and Darboux's Theorem are similar statements: Both assume a differentiable function on an interval, and show that there exists a point c in (a, b) where $f'(c) =$ something in particular.
How are these statements different? What is qualitatively different about the $f'(c)$ we're trying to find? Does one seem to imply the other, or are they unrelated?

Theorem 5.3.2 (Mean Value Theorem)

If $f : [a, b] \rightarrow \mathbb{R}$ is _____ on $[a, b]$ and _____ on (a, b) , then there exists a point $c \in (a, b)$ where

$$f'(c) =$$

Discuss your journal prompt, and draw a picture to explain.

Theorem 5.3.1 (Rolle's Theorem)

Let $f : [a, b] \rightarrow \mathbb{R}$ be _____ on $[a, b]$ and _____ on (a, b) . If $f(a) = f(b)$, then there exists a point $c \in (a, b)$ where $f'(c) = \underline{\hspace{2cm}}$.

Proof.

□

Now we can prove the Mean Value Theorem.

Scratch Work:

Proof.

□

Quick, powerful corollaries to the MVT:

Corollary 5.3.3

If $g : A \rightarrow \mathbb{R}$ is differentiable on an interval A and satisfies $g'(x) = 0$ for all $x \in A$, then g is a constant function.

Proof.

□

Corollary 5.3.4

If f and g are differentiable functions on an interval A , and $f'(x) = g'(x)$ for all $x \in A$, then

Antiderivatives differ by at most a constant.

Proof.

□

Theorem 5.3.5 (Generalized Mean Value Theorem/“Cauchy’s Mean Value Theorem”)

If f and g are continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b) , then there exists a point $c \in (a, b)$ where

$$[f(b) - f(a)]g'(c) = [g(b) - g(a)]f'(c)$$

What the hell even is this...

Proof.

□

After Class:

Pick & Prove Problems:

★★ Exercise 5.3.3

Let h be a differentiable function defined on the interval $[0, 3]$, and assume that $h(0) = 1$, $h(1) = 2$, and $h(3) = 2$.

- (a) Show that there exists a point $d \in [0, 3]$ where $h(d) = d$. Hint: See Exercise 4.5.7.
- (b) Show that at some point $c \in [0, 3]$, we have $h'(c) = 1/3$.
- (c) Show that $h'(x) = 1/4$ at some point in the domain.

★★ Exercise 5.3.7

A fixed point of a function f is a value x where $f(x) = x$. Show that if f is differentiable on an interval with $f'(x) \neq 1$ for any x , then f can have at most one fixed point.

Hint: Proceed by contradiction.

★★ Supplemental Exercise

Show that if $g : A \rightarrow \mathbb{R}$ is differentiable on an interval A and satisfies $g'(x) > 0$ for all $x \in A$, then $g(x)$ is increasing on A .

★★ Exercise 5.3.8

Assume f is continuous on an interval containing 0 and assume it's differentiable for all $x \neq 0$. If $\lim_{x \rightarrow 0} f'(x) = L$, show that $f'(0)$ exists and equals L .

Hint: Use the definition of the derivative.

Understanding Analysis Sections Covered: 5.2

Definitions and Theorems to include in Analysis Journal:

- Theorem 5.3.6 (L'Hospital's Rule, $0/0$ case)
- Theorem 5.3.8 (L'Hospital's Rule, ∞/∞ case)

Before Class:

Readings:

Analysis Journal Prompts:

- Write down L'Hospital's rule as well as you remember it from calculus. Once you've done that, now fill in L'Hospital's rule below as it's written in your textbook (Theorem 5.3.6). Is there anything new/different in this statement than the one you remember from calculus?

Theorem 5.3.6 (L'Hospital's Rule, $0/0$ case)

Let f and g be continuous on an interval containing a , and assume f and g are differentiable on this interval with the possible exception of the point a . If $f(a) = g(a) = 0$ and $g'(x) \neq 0$ for all $x \neq a$, then

Proof.

□

After Class:

Pick & Prove Problems:

★ Exercise 5.3.10

Let $f(x) = x \sin(1/x^4)e^{-1/x^2}$ and $g(x) = e^{-1/x^2}$.

Using any method, compute the limit as x approaches 0 of $f(x)$, $g(x)$, $f(x)/g(x)$, and $f'(x)/g'(x)$. Explain why the results are surprising but do not conflict with L'Hospital's Rule.

Theorem 5.3.8 (L'Hospital's Rule, ∞/∞ case)

Let f and g be continuous on (a, b) , and $g'(x) \neq 0$ for all $x \in (a, b)$. If $\lim_{x \rightarrow a} g(x) = \infty$ (or $-\infty$), then